Crop Recommendations with Ensemble Learning and Explainable AI

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Abstract. Crop recommendation systems are necessary for present day agriculture but most of them are unreliable due to their uninterpretability caused by small and single source databases. To rectify this concern we present a framework that combines ensemble learning and explainable AI. We have collected data from 4 different data repositories. We have then balanced the data sets using Synthetic Minority Over-sampling Technique (SMOTE). The data set contains 3,463 evidence-based samples across 26 crop classes. As a unique attribute we introduced CSI to capture Volatility in climate such as temperature, rainfall and humidity. Five models were evaluated under the rigorous 5-fold laminated cross-validation. Decision Tree 85.20% ± 0.61%, Random Forest 88.13% ± 0.80%, XGBoost 87.38% ± 0.57%, a Voting Ensemble of all three 86.55% ± 0.66%, as well as a hyperparameter-tuned Random Forest which provided the leading result 88.25% ± 0.51% (95% CI: 87.25%–89.25%) with the lowest variance across all models. SHAP-based post-hoc explanation relayed that rainfall, Nitrogen content and rainfall are the presiding factors. Overlapping feature distribution was observed in rice, wheat, barley, and sugarcane as they inhibit classification performance. This highlighted the necessity of local climate and micronutrient data in the future systems. The proposed framework demonstrated that comprehensibility and methodological soundness are achievable alongside veracity in agricultural decision support.

Keywords: Crop Recommendation, Ensemble Learning, Explainable Artificial Intelligence, SHAP, Random Forest, XGBoost, Climate Stress Index, Feature Engineering, SMOTE, Precision Agriculture

1 INTRODUCTION

Agriculture is the backbone of global food security but most of the time it is done without the guidance of analytics-driven investigation[2]. As the climate variability increases[3], the need for the recommendation systems have also increased. Most of these systems use the factors that affect agriculture the most, covering rainfall, nitrogen content, pH level and humidity[4]. Random Forest, decision tree and gradient boosting methods are some of the algorithms that have shown exceptional performance. There are 2 limitations that exist, first is that most of the systems are built on small data sets containing less than 2500 samples which leads to overfitting in accuracy, second is that most of the systems are black-boxed which limits its real world adoption [1].

SHAP (SHapley Additive exPlanations) is a theoretically based attributing feature for individual predictions[5]. It is not used in agriculture systems generally. Most of the

existing research papers which have included CSI into their study have focused on singular factors such as drought and rainfall[6]. But here this paper fills the void in the research. Contributions of this paper are a dataset of 3,463 samples of 26 crop classes unified across four public repositories that were expanded with SMOTE-balancing, a new feature of Climate Stress Index, a rigorous 5-fold cross-validation of five models including SHAP based global, individual class, and instance level breakdowns. This indicated that similar crops(agronomically similar like sugarcane, rice, wheat) lead to models being unable to tell them apart with the 7 basic features, N, P, K, temperature, humidity, pH, rainfall.

2 OBJECTIVES

The objective of this work is to develop an AI based crop recommendation system that supports informed agricultural decision making. The system does this by utilizing soil nutrient and climatic parameters. Through this study we aim to assess multiple ML models. This includes Decision Tree, Random Forest, XGBoost, Voting Ensemble, and a hyperparameter-tuned Random Forest under a rigorous 5 fold cross validation across a dataset of 3463 samples spanning 26 crops that was consolidated from multiple sources. Climate conscious feature engineering is integrated through Climate Stress Index (CSI). To improve transparency and comprehensibility to address limitations of the black box systems, Explainable Artificial Intelligence(XAI) techniques using SHAP are applied. This approach is designed to support sustainable agriculture and align with Sustainable Development Goal 2 (Zero Hunger).

3 LITERATURE REVIEW

3.1 Machine Learning Approach for crop recommendation

Soil nutrient and climatic parameters are used as inputs in ML based crop recommendation systems across various studies. In capturing non linear relationships between soil composition and crop suitability, Decision Tree, Random Forest, and XGBoost show strong performance in their classification[1][7]. Most crop recommendation system datasets suffer from overfitting and complex feature interaction, here Ensemble based approaches gained prominence due to their ability to handle these conditions[1][7]. As most studies were done with small datasets with less than 2500 samples, this resulted in them having concerns about generalization and accuracy of the reported results.

3.2 Limitations of Existing ML-Based Approaches

A repeated limitation of the studies in the Agricultural field is that they are treating the climate variable as individual instances without taking into account their combined variance from the crop's optimal range. This stops the model from directly capturing multifactorial stress combinations that directly affect the crops sustainability[8]. Furthermore, existing systems are evaluated on the accuracy matrics

and are not cross validated which results in diminishing credibility of the reported results[9].

3.3 Ensemble and boosting techniques in agriculture Prediction

Due to the improved stability and generalization over single classifiers of the ensemble models, models such as decision tree and XG Boost are adopted into recommendation systems[7]. Yield prediction and classification of crops have become consistent through ensemble voting and boosting[10]. However, most of these models lack transparent internal decision making behaviours or feature contribution which limits the adoption of the model in real time.

3.4 Explainable Artificial Intelligence in Agriculture

Explainable AI has emerged as an important component of black box models. SHAP and LIME have been applied in yield prediction and crop disease detection to quantify feature contributions at the instance level[11]. Despite showing practicality in similar domains, SHAP based clarification remains underutilized in this field[11]. The limiting factor of the crop recommendation adoption in agriculture stems from the lack of explainability of why a crop was recommended.

3.5 Research Gaps Identified in Existing Literature

The literature reveals three major gaps in the previous studies. First being the single source databases without rigorous validation. Second being the lack of climate aware features that can capture composite stress conditions. Lastly the lack of interpretability of results on the instance level which decreases the trust of the users. Here in this paper we address these three gaps by consolidating datasets from multiple public repositories, a CSI and a SHAP based explanations. This framework supports the Sustainable Development Goal 2 (Zero Hunger)[12].

4 METHODOLOGY

Data preparation, feature engineering, model training, ensemble learning, and explainability analysis is the structured pipeline for the proposed methodology. The objective of the study is to develop an easy to understand and climate conscious crop recommendation.

4.1 Data Collection and Preprocessing

We consolidated four different public datasets into a single and unified dataset to address the limited scale of a single source database. Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall were the seven main agronomic parameters. It was made sure that the dataset's target variable was the crop label. Raw datasets were standardised through column name normalisation, conversion of mixed-type fields to numeric. Null values were also removed. The duplicates were removed from the datasets by comparing and matching all the all seven features and the crop label. The non ecologically valid features were removed by the 1st–99th percentile per-feature filter. This reduced the outliers in the dataset. After all the cleaning, the dataset contains 3,463 samples across 26 crop classes in comparison to the 2,200 or less single source dataset used in other studies.

4.2 Class Balancing via SMOTE

Each crop contains sample counts ranging from 81 to 261 per class. This showed moderate imbalance across 26 crop classes in this cleaned and consolidated dataset. . Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training portion to address this issue. This generated synthetic and artificial samples through k-nearest neighbour interpolation (k=5) in feature space. The test set was not involved in the dataset when the SMOTE application was applied to the training dataset. This makes the test dataset contain only real and legitimate samples, this ensures legitimacy of the accuracy and results. Due to dataset being artificially inflated with SMOTE now all the crop classes contain 209 training samples, this took the total count of the dataset from 3,463 to 5434 samples.

4.3 Feature Engineering

Originally there were seven base soil and climate features. Beyond that we have engineered 15 features to improve crop class categorization. This was done for the chemically similar cereal crops. N/P ratio, N/K ratio, P/K ratio, total NPK, and NPK balance were the nutrient ratio features. Crop families can be differentiated from each other based on the dominance of nutrients in the soil. Temperature × humidity, rainfall × humidity, and temperature × rainfall are the climate interaction terms that were introduced to encode combined climatic effects. pH sensitivity bands that is, acidic: pH < 6.0, neutral: 6.0 – 7.5, alkaline: pH > 7.5were introduced along with rainfall categories low: < 60mm, medium: 60 – 150mm, high: > 150mm were encoded as binary indicator features.

A Climate Stress Index (CSI) was introduced to capture combined climate deviation from mean conditions. These are computed as:

$$\mathbf{CSI = 0.4|T| + 0.3|H| + 0.3|R|}$$

where T, H, and R represent the standard zero-mean and unit-variance values of temperature, humidity, and rainfall respectively. The weighting coefficients are $\alpha=0.4$, $\beta=0.3$, $\gamma=0.3$. These assign greater sensitivity to temperature deviation, thus reflecting its relatively stronger influence on crop suitability. The dataset finally consists of 22 features used for training the model.

4.4 Model Training and Testing

Decision Tree(DT), Random Forest(RF), XGBoost, a soft voting ensemble of all three and a hyperparameter tuned Random Forest were the five classifiers were trained and evaluated. RandomizedSearchCV with 30 iterations over a grid spanning n_estimators, max_depth, max_features, min_samples_split, min_samples_leaf, and class_weight, using 5-fold stratified cross-validation as the inner loop, the tuned RF configuration was identified.

All the models were evaluated under the 5-fold stratified cross-validation on the SMOTE enhanced training set and the real test set. Thus reporting mean accuracy, standard deviation, and 95% confidence intervals. This replaces the usual 80/20 training and testing evaluation used in prior studies that involved crop recommendation. This provided statistically reliable performance estimates. Accuracy, Precision, Recall, and F1-Score are the four evaluation metrics. They are computed as:

$$\textbf{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\textbf{Precision} = TP / (TP + FP)$$

$$\textbf{Recall} = TP / (TP + FN)$$

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$$

4.5 SHAP-Based Explainability

The tuned Random Forest provided a multi level model interpretation when SHAP (SHapley Additive exPlanations) was applied. A Shapley value is assigned to each feature, this represents its marginal contributions to a given prediction. These are computed as:

$$\varphi_i = \Sigma \, [\, |S|! \, (\, |F| - |S| - 1 \,)! \, / \, |F|! \,] \times [\, f \, (\, S \cup \{i\} \,) - f \, (S) \,]$$

where F is the full feature set, S is a feature subset excluding feature i, and f(S) denotes model output for subset S. This generated three levels of explanations. Firstly, feature importance across all 26 classes are ranked on a global mean |SHAP| bar chart. Secondly, each feature's impact across the test set is graphed on a beeswarm plot showing the direction and magnitude. Lastly, the six most confusing crop classes, barley, wheat, rice, sugarcane, soybean, and cotton are shown on the per class SHAP bar charts, for the six most confused crop classes. This reveals which features the model relies on when discrimination is most difficult. To provide concrete prediction level explanation to farmer decision support, instance level waterfall plots were generated for a correctly predicted rice sample and a misclassified soybean sample.

5 SYSTEM ARCHITECTURE

This study has proposed a system that follows a modular six-layer pipeline. Here each of the layered components have a specific function. This ensures that the system is reproducible and scalable with transparency. The complete workflow is illustrated in Fig. 1.

5.1 Data Input Layer

This part of the architecture/pipeline uses the seven standard agronomic parameters, Nitrogen(N), Phosphorus(P), Potassium(K), temperature, humidity, pH, and rainfall. These parameters can be from a public repository, IoT based soil sensors, meteorological stations or direct farmer-provided inputs. This ensures the system remains adaptable across deployment contexts.

5.2 Preprocessing Layer

Raw inputs undergo column standardisation, numeric type validation, null removal, exact duplicate elimination and outlier filtering. We use 1st–99th percentile per feature threshold. SMOTE is applied to correct the class imbalance in multi-source data. The SMOTE is applied to the training partition only. As the test set does not go through SMOTE, so the integrity of the test set remains.

5.3 Feature Engineering Layer

The total number of preprocessed features have been extended from 7 to 22. There are three categories of engineered features. Nutrient ratio features N/P, N/K, P/K, total NPK, NPK balance. Climate interaction terms temperature × humidity, rainfall × humidity, temperature × rainfall. And indicator features that show threshold for pH bands and rainfall categories. By aggregating normalised deviations of humidity, temperature and rainfall, we have converted it into a single composite feature. This is called the Climate Stress Index (CSI). Rather than treating the variables independently, the model that has been developed will treat and capture the combined stress of the climate.

5.4 Model Training Layer

The features are passed through the classifiers. These include Decision Tree, Random Forest, XGBoost, a soft voting ensemble of all three and hyperparameter tuned Random Forest. The hyperparameter tuned Random Forest was identified through RandomizedSearchCV. All of the models were trained under the SMOTE, i.e. the balanced training model. And then are evaluated under 5-fold stratified cross-validation. The report consists of mean accuracy with Standard Deviation and the 95% confidence intervals.

5.5 Ensemble and Optimisation Layer

The probability is combined from the outputs of Decision Tree, Random Forest and XGBoost. This is done by the soft voting ensemble. Here the crop class with the highest aggregated probability is selected. The Random Forest was fine-turned separately by testing the combinations of the six settings. This made the model very stable and consistent with the results. The results were the most stable and reliable in the five models. The confidence interval of this model was the lowest being ±0.51%.

5.6 Explainability Layer

Three levels of explanations are generated from the SHAP that was applied to the tuned Random Forest. These explanations are global feature importance rankings across all 26 crop classes, per-class importance charts for the six most confused crops and instance-level waterfall plots for individual predictions. Because of this layer the explanations of the decision will be transparent to the farmers and decision makers. The transparency will justify and explain the reason for the recommendation of a certain crop for a soil composition and the climate.

5.7 Output Layer

The system provides the best crop recommendation lead by confidence score and SHAP based feature attribute. It also identifies which input parameter would be more applicable in the prediction. It is designed so that the output is interpretable to non-technical members while the remaining rigorous agricultural researchers and policy applications.

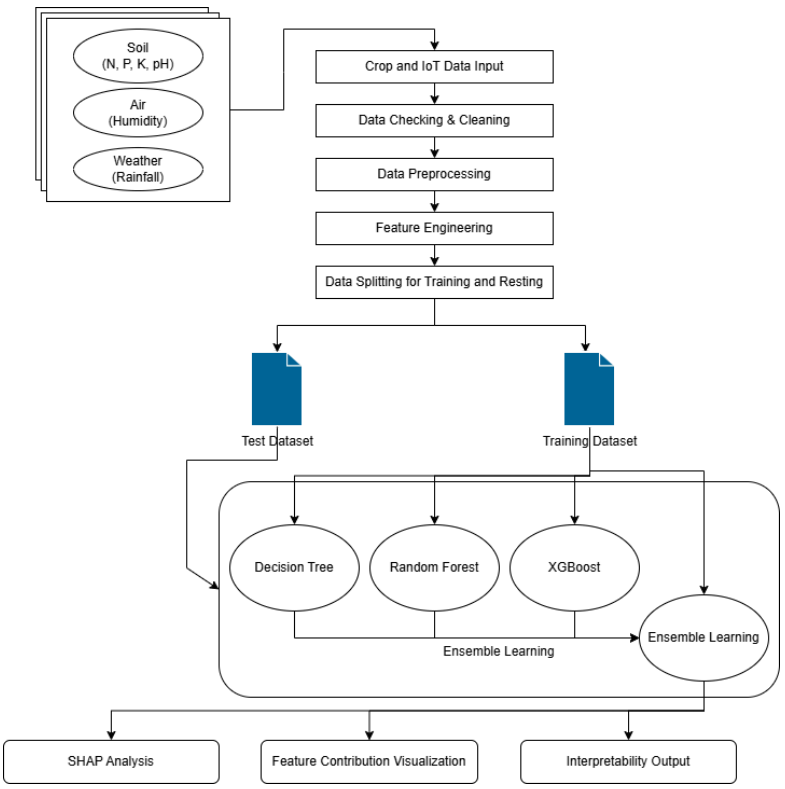

Fig. 1. Proposed System Architecture.

6 RESULTS

6.1 Model Performance Comparison

Table 1 displays the performance of the five classifiers. These were evaluated on the test set of 693 real samples that were left out alongside with 5-fold stratified cross validation results. The models used in this study are trained in the bases of the SMOTE balance training set which has a set of 5,434 samples all across 26 crop classes. This is shown in Fig. 2.

TABLE I: MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score	CV Accuracy (Mean ± Std)	95% CI
Decision Tree	0.795	0.793	0.795	0.793	0.8520 ± 0.0061	0.840 – 0.864
Random Forest	0.798	0.812	0.798	0.803	0.8813 ± 0.0080	0.866 – 0.897
XGBoost	0.797	0.801	0.797	0.798	0.8738 ± 0.0057	0.863 – 0.885
Tuned RF (proposed)	0.795	0.812	0.795	0.801	0.8825 ± 0.0051	0.873 – 0.893
Ensemble (RF+DT+XGB)	**0.805**	**0.802**	**0.805**	**0.802**	0.8655 ± 0.0066	0.853 – 0.878

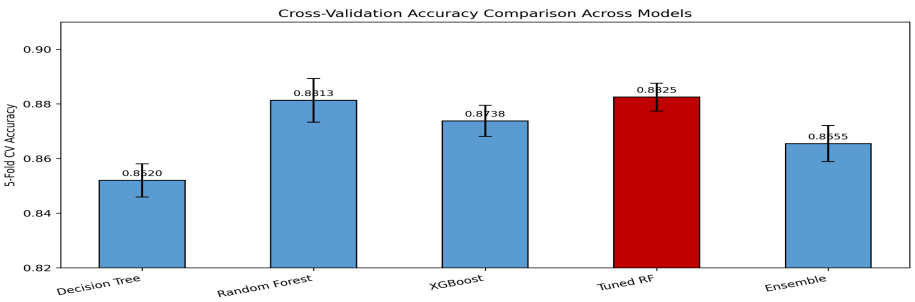

Fig. 2 Cross-validation accuracy comparison across all five models with error bars representing standard deviation

Test set accuracy of the Ensemble was the highest at 80.5%. The Highest cross validation is at 88.25% that was achieved by Random Forest with the lowest standard deviation of 0.51%. This indicates that across all folds it gives the most consistent generalization performance.

6.2 **Per-Class Performance**

Of the 26 crop classes that have achieved F1 score of 0.96 or more also includes apple, banana, blackgram, chickpea, coconut, grapes, kidney beans, lentil, mango, muskmelon, orange, papaya, jute, coffee, pomegranate and watermelon. There are 6 crops that show difficulty in classification, this persistence follows throughout the study. These are barley(f1:0.111), soybeans(F1:0.185), wheat(F1:0.170), sugarcane(F1:0.169), rice(F1:0.556), cotton(F1:0.645). This is shown in Fig. 3.

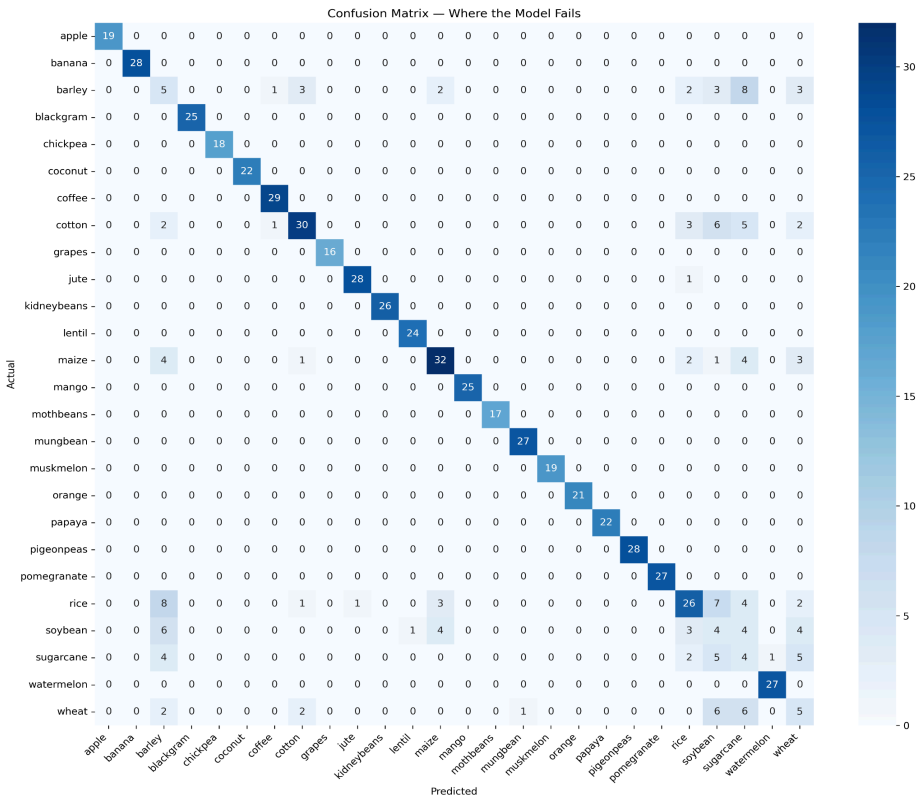

Fig. 3 Confusion matrix of the Ensemble model on the held-out test set of 693 samples

6.3 SHAP Feature Importance

Analysis from Global SHAP shows rainfall, nitrogen and temperature as the three most influential features of all over 26 classes. This is shown in Fig. 4. The more rainfall values strongly drive prediction on water intensive crops in Bee swarm (Fig. 5) while high nitrogen is associated with cereal crop recommendation. The reason for the low F1 scores of the six cereal crops is due to their near identical features, this was found by Per-class SHAP analysis. This is shown in Fig. 6. Instance-level waterfall analysis for a correctly predicted rice sample shows high rainfall as the dominant positive contributor. Whereas the near zero humidity and rainfall of the samples in soybean in SHAP values caused the model to default to the wheat prediction based on nitrogen and potassium alone.

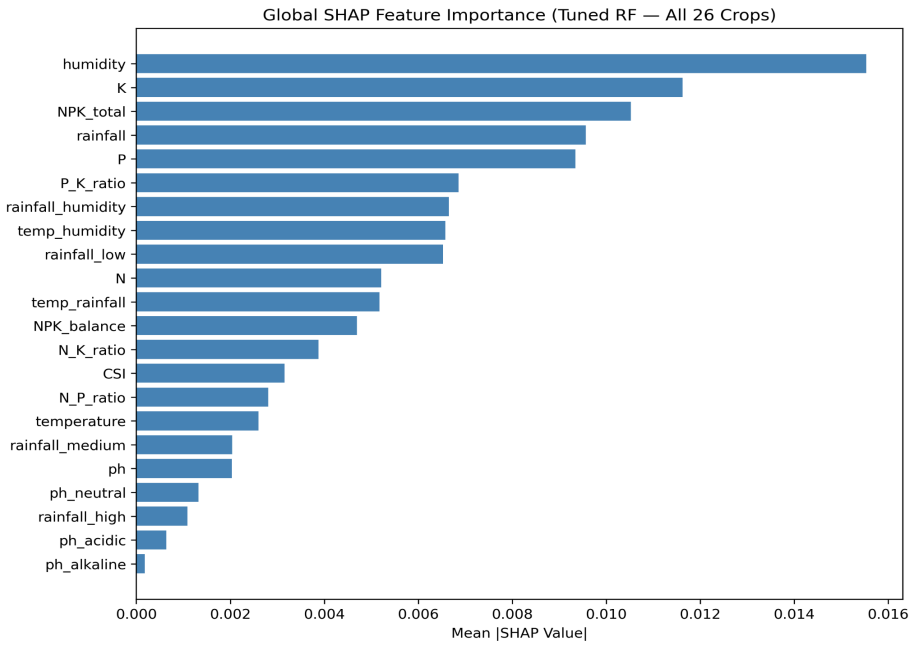

Fig.4 Global SHAP mean absolute feature importance across all 26 crop classes

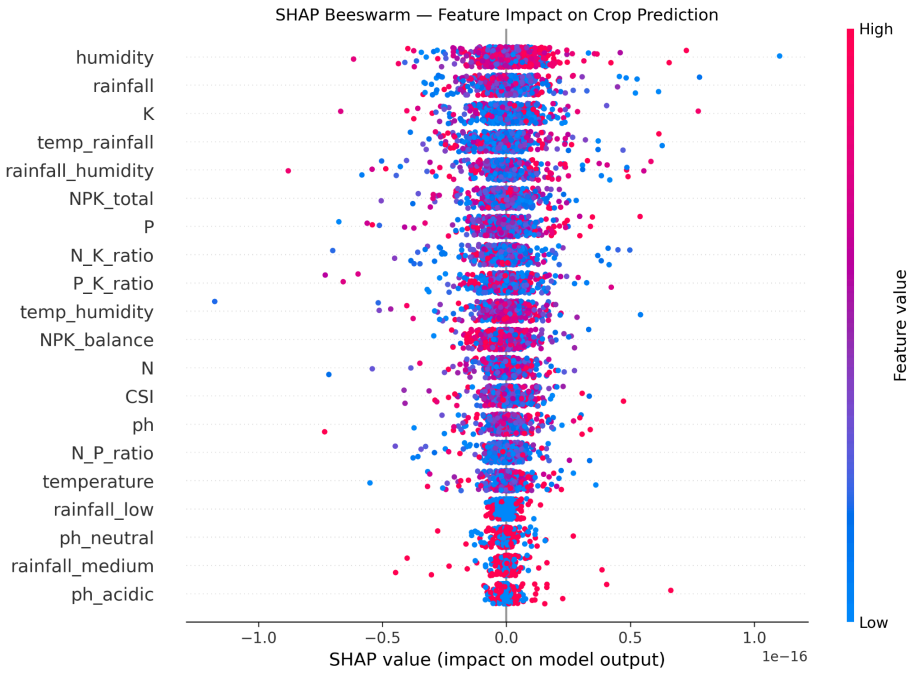

Fig. 5 SHAP beeswarm plot showing direction and magnitude of feature impact across the test set

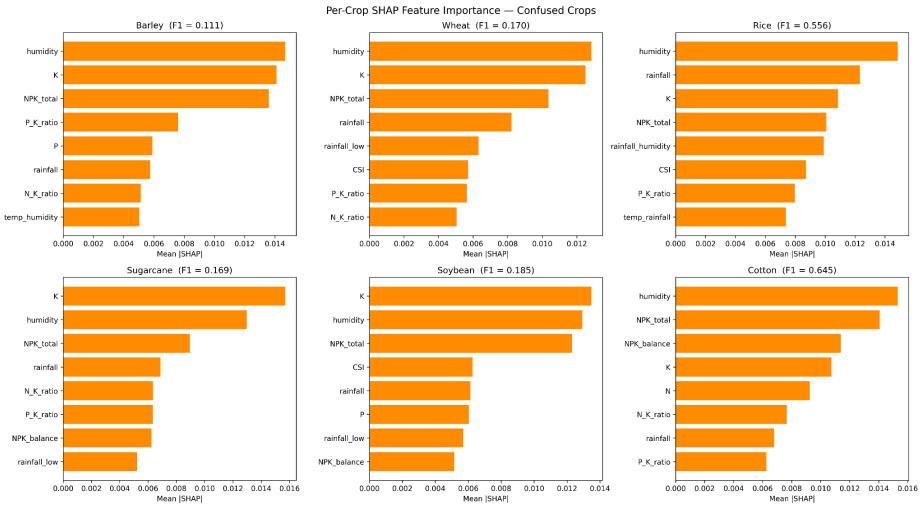

Fig. 6 Per-class SHAP feature importance for the six confused crop classes with corresponding F1-scores

7 DISCUSSIONS

The results have shown that the crop recommended system can reliably recommend crops for 18 of 26 classes with almost no errors. This validated the effectiveness of using the multisource data and then consolidating them into a single dataset, SMOTE balancing and the features that were artificially engineered for the chemically and agronomically distinct crop. 88.25% ± 0.51% was the best cross validation stability reached by the Tuned Random Forest. The Ensemble shows great strength of the combined Decision Tree, Random Forest, and XGBoost through soft voting as it closely leads on raw test accuracy of 80.5%.

The continuous misclassification of barley, wheat, rice, sugarcane, soybean, and cotton defines the most significant finding of this study. Through SHAP analysis these crops share nearly indistinguishable patterns in feature attributes in rainfall, temperature and nitrogen. This makes the removal of the discrimination of these crops in the confident model with the present seven parameters nearly impossible. This does not mean that this is a mistake in the creation of the model but an issue with the limitation in data. There is no way to separate the features of these crops without adding additional discriminating information as most of these crops are genuinely grown in overlapping South Asian climate zones with similar soil nutrient requirements.

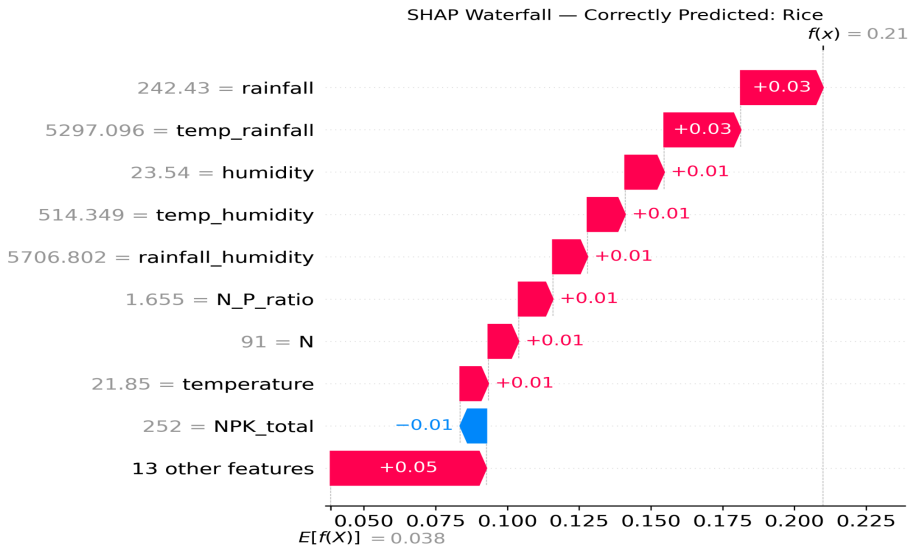

Fig. 7 SHAP waterfall plot for a correctly predicted rice sample showing dominant feature contributions

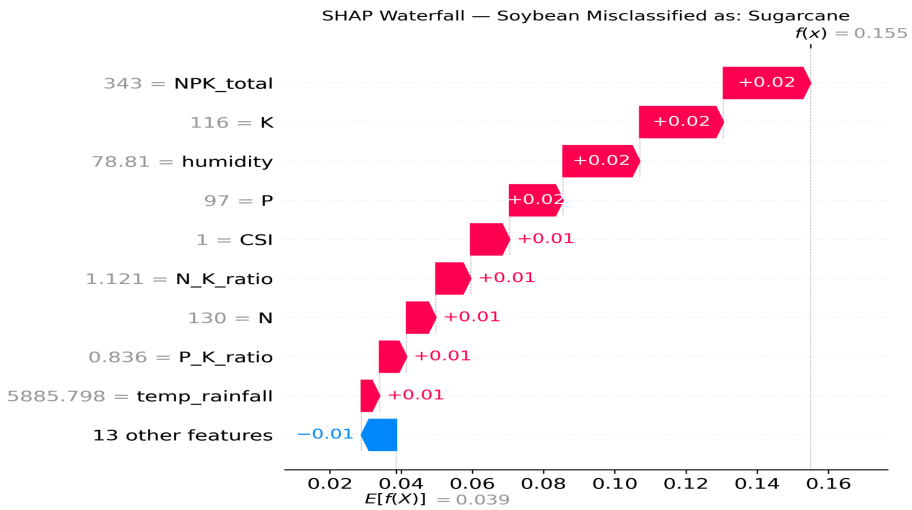

Fig. 8 SHAP waterfall plot for a misclassified soybean sample predicted as wheat, illustrating feature overlap

The future datasets design will have direct implications from this finding. Due to the limited seven parameters in the dataset the models can not distinguish between the six crops, but if future studies include the soils trace micronutrients, a comprehensive climate data with the soil physical properties it will be possible to distinguish between them. The SHAP charts that were generated in this study does not just explain the model's decision but also tells which parameters should be added to the dataset for future studies. Through this it can be seen that explainable AI such as SHAP are more than just explanations for the farmers and policy makers but also help future researchers make improved datasets.

The Climate Stress Index, helped the model but could not overcome the confusion built by the 6 similar crops. As these crops grow in similar climate conditions they could not overcome this confusion with CSI as it ultimately looked similar to the other 6. In the future experiment, researchers should focus on crop specific parameters with independent CSI for each crop. This can be achieved by focusing on real world agricultural experiments rather than mathematical optimization. This will make sure the difference between the rainfall sensitive rice and temperature sensitive wheat can be distinguished.

VII. CONCLUSION

This paper presented a climate-aware, interpretable crop recommendation framework combining ensemble machine learning with SHAP-based Explainable AI. By merging 4 public databases, consolidating them into a single dataset of 3,463 samples across 26 crop classes. This was followed by removal of duplicates, elimination of outliers and SMOTE based class balancing. From the seven original agronomic parameters

present in the original datasets, twenty two features were engineered with CSI to capture the composite deviation climate deviation across rainfall, temperature and humidity

Five classifiers were evaluated under 5-fold stratified cross-validation. The Tuned Random Forest achieved the best generalisation stability at 88.25% ± 0.51% (95% CI: 87.25%–89.25%), while the Voting Ensemble achieved the highest test set accuracy at 80.5%. Eighteen of 26 crop classes were predicted with F1-scores of 0.96 or above. SHAP analysis identified rainfall, nitrogen, and temperature as the globally dominant features, while per-class explanation revealed that six climatically similar cereal crops — barley, wheat, rice, sugarcane, soybean, and cotton — share near-identical feature attribution patterns, constraining reliable discrimination with standard agronomic inputs alone.

These findings contribute on two levels. Technically, the framework demonstrates that statistically rigorous evaluation, multi-source data consolidation, and instance-level explainability are simultaneously achievable in agricultural recommendation systems. Practically, the SHAP per-class analysis provides a data-driven basis for identifying which additional measurements — micronutrients, regional climate indices, soil texture — would most improve future systems. This work aligns with Sustainable Development Goal 2 (Zero Hunger) by advancing transparent, trustworthy AI tools for agricultural decision support.

The upcoming research will tell us three areas which include using micronutrient features and developing crop specific CSI model with validated coefficient and testing the model on agriculture data from outside South Asia to enhance its performance across different farming regions.

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